

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions



Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data dimensions tracked

37

Missing Values

Review ratings imputed

Customer demographics, purchase details, and shopping behavior captured across product categories, seasons, and locations.

Python Data Preparation

Data Loading & Exploration

- Imported dataset using pandas
- Checked structure with `df.info()`
- Generated summary statistics

Data Cleaning

- Imputed missing review ratings
- Standardized column names
- Dropped redundant fields



Feature Engineering



Age Groups

Binned customer ages into segments



Purchase Frequency

Calculated days between purchases



Database Integration

Loaded to PostgreSQL for SQL analysis

SQL Analysis: Revenue Insights

Revenue by Gender

- **Male:** \$157,890
- **Female:** \$75,191

Shipping Comparison

- **Standard:** \$58.46 avg
- **Express:** \$60.48 avg

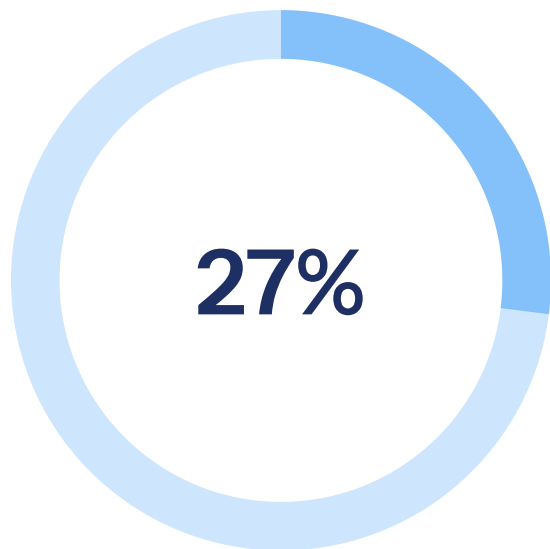
Discount Users

839 customers used discounts while spending above average purchase amount.

Top Products by Rating

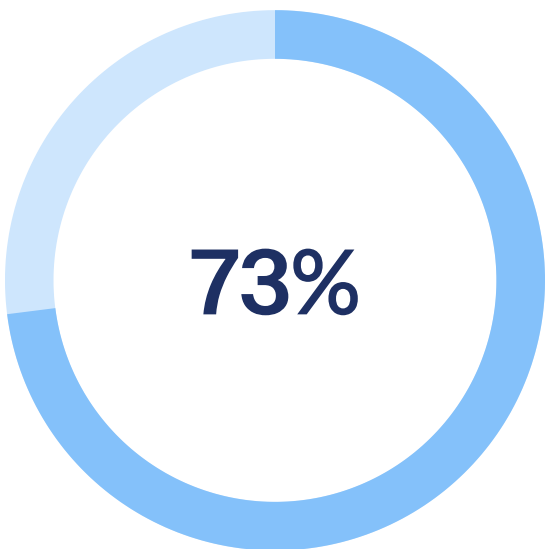
1. Gloves (3.86)
2. Sandals (3.84)
3. Boots (3.82)

Subscription Analysis



Subscribers

1,053 customers subscribed



Non-Subscribers

2,847 customers not subscribed

Average Spend

Subscribers: \$59.49
Non-Subscribers: \$59.87

Total Revenue

Subscribers: \$62,645
Non-Subscribers: \$170,436

Customer Segmentation



Loyal

3,116 customers with consistent purchase history



Returning

701 customers making repeat purchases



New

83 customers with single transactions



Product Performance

Top Products by Category

- **Clothing:** Blouse, Pants, Shirt
- **Accessories:** Jewelry, Sunglasses, Belt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Discount-Dependent Products

1. Hat (50%)
2. Sneakers (49.66%)
3. Coat (49.07%)



Power BI Dashboard

Revenue by Category

Clothing (\$100K) leads, followed by Accessories (\$70K), Footwear (\$30K), Outerwear (\$20K)

Revenue by Age Group

Young Adult (\$45K) highest, Middle-aged (\$40K), Adult (\$35K), Senior (\$30K)

Sales Volume

Clothing (1,800 sales) dominates, Accessories (1,200), Footwear (500), Outerwear (200)

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base



Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment



Review Discount Policy

Balance sales boosts with margin control

Targeted Marketing

Focus on high-revenue age groups and express-shipping users